

Andrew Lugo

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EDUCATION

New Jersey Institute of Technology
Bachelor of Science in Computer Science

Newark, NJ
August 2021 – May 2025

WORK EXPERIENCE

Backend Software Engineer

January 2025 – May 2025

TheoForge Platform NJIT | Python, FastAPI, Neo4j, OpenAI, PostgreSQL

Newark, NJ

- **AI-Powered Knowledge Discovery Platform:** Built and scaled distributed backend services using **FastAPI**, **PostgreSQL**, **Neo4j**, and **Qdrant**, reducing average query response time by **35–45%** through optimized data retrieval and indexing strategies.
- **REST API & Authentication System:** Developed **15+ RESTful API endpoints** for user management, transcript processing, and knowledge graph generation using **FastAPI**, implementing **JWT authentication** and **Dockerized deployment** to support secure and consistent service access.
- **Backend Infrastructure & Data Systems:** Engineered storage and database infrastructure using **MinIO**, **PostgreSQL**, and **Neo4j/Qdrant**, managing **thousands of structured and vectorized records** and improving data access efficiency across services.
- **Natural Language Query System (LLM Integration):** Integrated **OpenAI APIs** with **FastAPI** to enable natural language querying over structured knowledge graphs, reducing manual data lookup effort by **30%** and improving usability for non-technical users.
- **CI/CD & Development Workflow:** Implemented **CI/CD pipelines** using **GitHub Actions**, reducing deployment time by **40%** and improving development iteration speed within Agile workflows using Jira.

Technical Operations Assistant (Software Development)

June 2025 – Present

Jag & Son Construction | WordPress, HTML

Kearny, NJ

- **Customer Inquiry Automation System:** Developed and customized a WordPress contact form using **HTML** and **WordPress**, reducing manual scheduling effort by **40%**.
- **Operations Workflow Management:** Coordinated customer scheduling and contract processing, improving response time and streamlining client intake and appointment workflows.

PROJECTS

Coup Game CoveyTown | *AWS EC2, Typescript, Node.js, React*

- **Event-Driven Backend System:** Engineered **event-driven backend services** using **Node.js** and **TypeScript** to process real-time player actions and synchronize distributed game state across clients, supporting **50+ concurrent simulated players** during testing.
- **Real-Time State Management:** Designed server-side **state management** logic to handle concurrent gameplay events and maintain consistent game state across multiple clients.
- **Server-Authoritative Architecture:** Implemented **server-authoritative** game logic to ensure consistent state across clients, reducing desynchronization during real-time multiplayer interactions.
- **Low-Latency Event Processing:** Optimized event handling and state synchronization logic, reducing **update latency** and ensuring responsive real-time gameplay during **concurrent player interactions**.

PeakFit Workout Tracker | *Kotlin, Firebase Firestore, Firebase Auth*

- **Real-Time Data Synchronization System:** Engineered real-time data synchronization using **Firebase Firestore** and **Auth**, enabling consistent cross-device state and handling **hundreds of read/write operations** per session.
- **Backend Data Modeling & Query Optimization:** Designed and optimized **Firestore** data models and queries, reducing data access latency by **25%** and improving performance of workout tracking operations.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, JavaScript, TypeScript, SQL, Bash, C/C++

Frameworks & Libraries : FastAPI, Node.js, Express, React, REST APIs

Databases & Storage : PostgreSQL, MongoDB, Neo4j, Qdrant, Firebase Firestore, MinIO

Tools & Platforms : Docker, AWS EC2, GitHub, GitHub Actions, Firebase, pgAdmin, Jira, Alembic, JWT

Other: HTML/CSS, MATLAB